THOMAS L. FRIEDMAN

Hot, Flat, and Crowded

WHY WE NEED A GREEN REVOLUTION—AND HOW IT CAN RENEW AMERICA

RELEASE 2.0

UPDATED AND EXPANDED

PICADOR / FARRAR, STRAUS AND GIROUX

NEW YORK

As we get rising global average temperatures and the earth gets warmer, it will trigger more evaporation from the soil. So regions that are already naturally dry will tend to get drier. At the same time, higher rates of evaporation, because of global warming, will put more water vapor into the atmosphere, and so areas that are either near large bodies of water or in places where atmospheric dynamics already favor higher rates of precipitation will tend to get wetter. We know one thing about the hydrologic cycle: What moisture goes up must come down, and where more moisture goes up, more will come down. Total global precipitation will probably increase, and the amount that will come down in any one storm is expected to increase as well—which will increase flooding and gully washers. That's why this rather gentle term "global warming" doesn't capture the disruptive potential of what lies ahead.

"The popular term 'global warming' is a misnomer," says John Holdren. "It implies something uniform, gradual, mainly about temperature, and quite possibly benign. What is happening to global climate is none of those. It is uneven geographically. It is rapid compared to ordinary historic rates of climatic change, as well as rapid compared to the adjustment times of ecosystems and human society. It is affecting a wide array of critically important climatic phenomena besides temperature, including precipitation, humidity, soil moisture, atmospheric circulation patterns, storms, snow and ice cover, and ocean currents and upwellings. And its effects on human well-being are and undoubtedly will remain far more negative than positive. A more accurate, albeit more cumbersome, label than 'global warming' is 'global climatic disruption.'"

CNN.com reported (August 7, 2007) about a study that had just been released by the UN's World Meteorological Organization of unprecedented weather extremes that had happened so far that year. It could have just been called "Global Weirding 2007":

> Four monsoon depressions, double the normal number, caused heavy flooding in India, Pakistan and Bangladesh . . . England and Wales have experienced their wettest May-to-July period since record-keeping started in 1766. In late July, swollen rivers threatened to burst their banks . . . Late last month in Sudan, floods and heavy rain caused 23,000 mud brick homes to collapse, killing at least 62 people. The rainfall was abnormally heavy and early for this time of the year . . . In May, swell waves up to 15 feet high swept into 68 islands in the Maldives, causing severe flooding and damage . . . Also in May, a heat wave swept across Russia . . . Southeastern Europe did not escape the unusual weather. The area suffered record-breaking heat in June and July . . . An unusual cold southern winter brought wind, blizzards and rare snowfall to various parts of South America, with temperatures reaching as low as 7 degrees below zero Fahrenheit (–22 degrees Celsius) in Argentina and 0 degrees Fahrenheit (–18 degrees Celsius) in Chile in July. In June, South Africa had its first significant snowfall since 1981, as almost 10 inches (25 centimeters) of the white stuff fell in some parts of the country . . .

This trend of more extreme extremes continued right into the summer of 2008, when unprecedentedly heavy rains in Iowa caused the Cedar River to flood and overwhelm downtown Cedar Rapids, rising well above thirty feet over sea level—far, far above what anyone had ever seen or expected. A quote in the *New York Times* report (June 13, 2008) captured perfectly the sense of global weirding that Iowans felt: "'Usually if you break a record, you only do it by an inch or two,' said Jeff Zogg, a hydrologist for the Weather Service in Davenport, Iowa. 'But breaking it by six feet? That's pretty amazing.'"

We're Not in Kansas Anymore

So where are we, then, as we enter the Energy-Climate Era? Where we are is that many people think the weather has gotten strange, but not enough think it has gotten so dangerous that we need to do what is necessary to manage the unavoidable and avoid the unmanageable. We need to stretch our imaginations. We need to understand that boundaries of change are far wider than the average of projections.

One of the things Heidi Cullen, formerly the Weather Channel's climatologist, has argued—in order to stimulate people's understanding and imagination regarding climate change—is that local TV meteorologists should give climate-change science some mention in their daily weather reports. "We owe it to viewers to connect climate to weather when there is a meaningful connection in terms of broad trends," said Cullen, who holds a Ph.D. in climatology and ocean-atmosphere dynamics from the